SBI4FINANCE / NUMERICAL EVIDENCE / 28 OCTOBER 2026

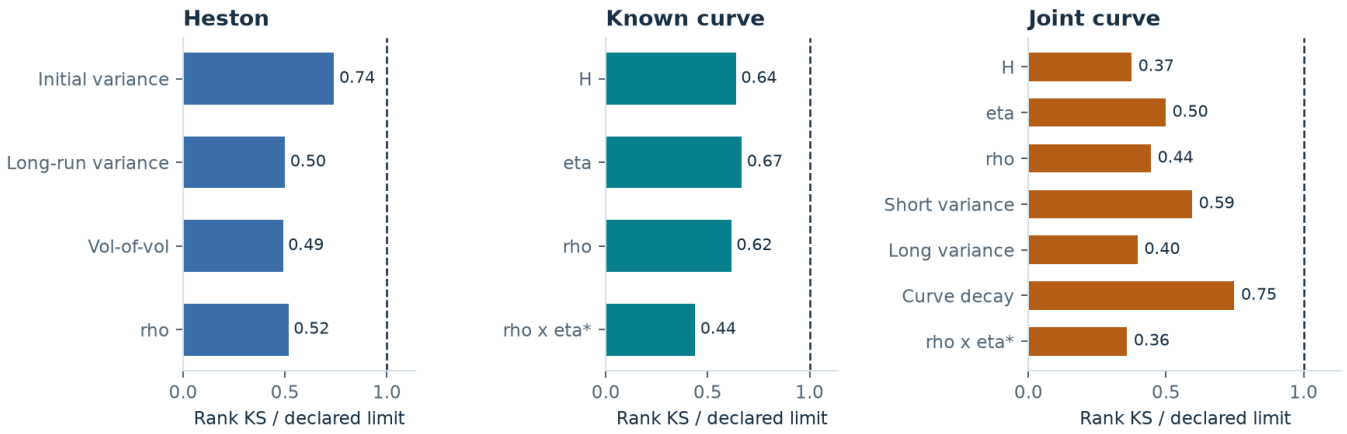
Inference performance

A readable review of the completed tutorial fits. Three baseline models are shown throughout; page 8 adds the newly completed known-curve 100k fit and its matched learning comparison.

Trained model	Training surfaces	Evaluation panel	Rank screens
Heston / final epoch 500	100,000	2,000 held-out test cases	4 / 4 pass
Rough Bergomi / known curve	50,000	2,000 validation cases	4 / 4 pass*
Rough Bergomi / inferred curve	20,000	2,000 validation cases	7 / 7 pass*
Known curve / later fit (page 8)	100,000	Same 2,000 validation cases	4 / 4 pass*

Calibration screens are encouraging for the baseline and later fits. Heston has the strongest independent reference evidence. Known-curve rough inference matches two of three approximate emulator-PyMC cases; full model-posterior, pricing and product acceptance remain unfinished.

Baseline rank calibration: every recorded target



Bars show rank-KS distance divided by each model's declared simultaneous limit. Values below the dashed line pass that model's screen. Limits differ with the number of targets. [S1-S3]

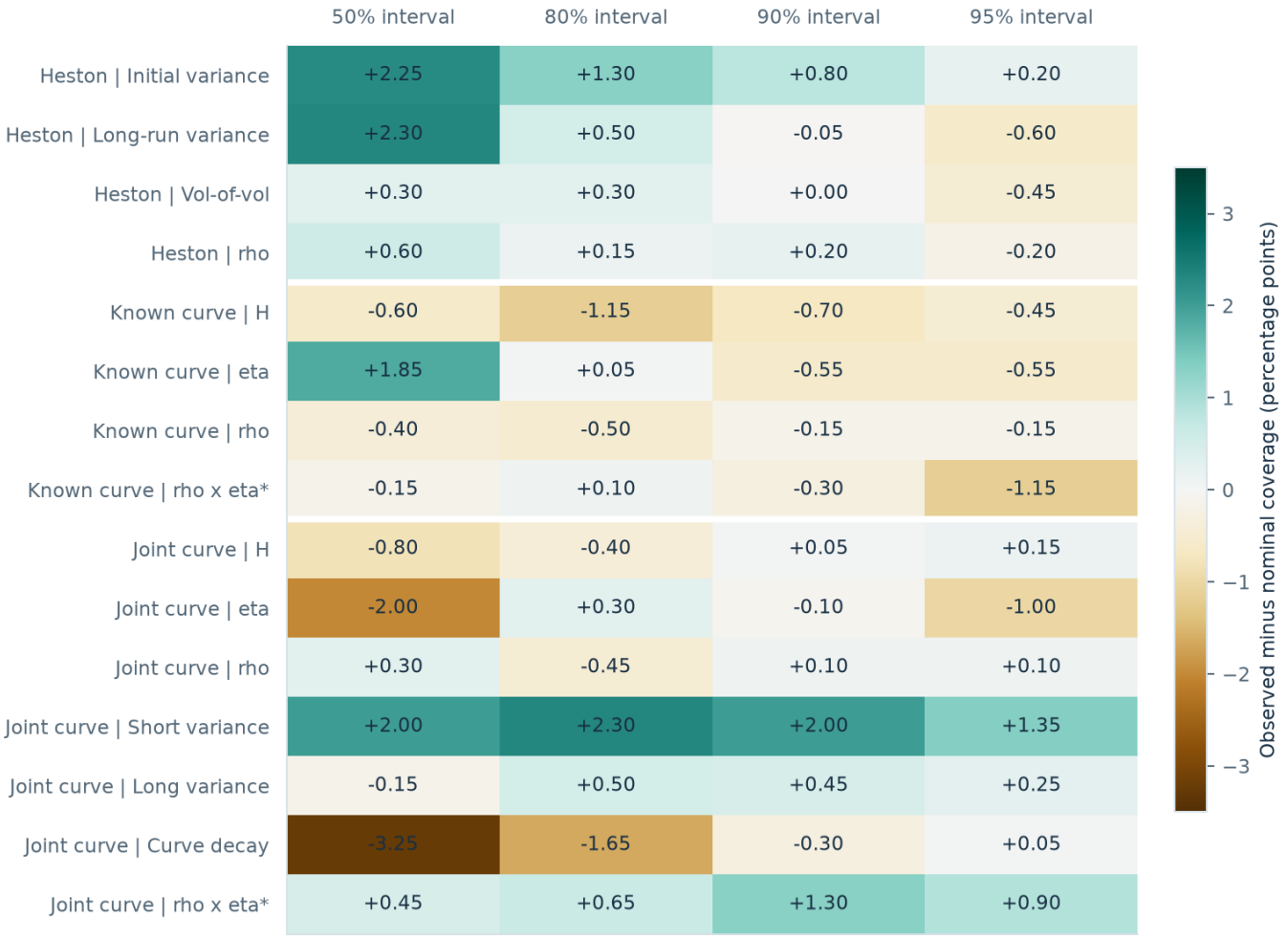
* rho x eta is computed downstream from each joint posterior draw; it is not another inferred parameter. Heston has four inferred parameters with kappa fixed at 2; the known-curve rough model has three, and the joint-curve rough model has six.

The figures on pages 1-7 retain Heston 100k, known-curve 50k and joint-curve 20k as explicit baselines. The completed known-curve 100k fit is additional development evidence, without cache promotion; its inference and price emulator are assessed separately on page 8. The joint 100k bank extension remains separate.

02 / UNCERTAINTY CALIBRATION

Passing ranks is a starting point

Every panel uses 2,000 simulated observations and 1,000 posterior draws per observation. Heston's candidate and rules were frozen before opening these test outcomes. Both rough panels are development validation; their reserved tests remain unused for inference evaluation.



Cell values are percentage-point deviations from nominal central-interval coverage. Brown is undercoverage; green is overcoverage. rho x eta* uses aligned joint samples. These are separate model-wise diagnostics, not a cross-model test of superiority. [S1-S3]

What requires attention

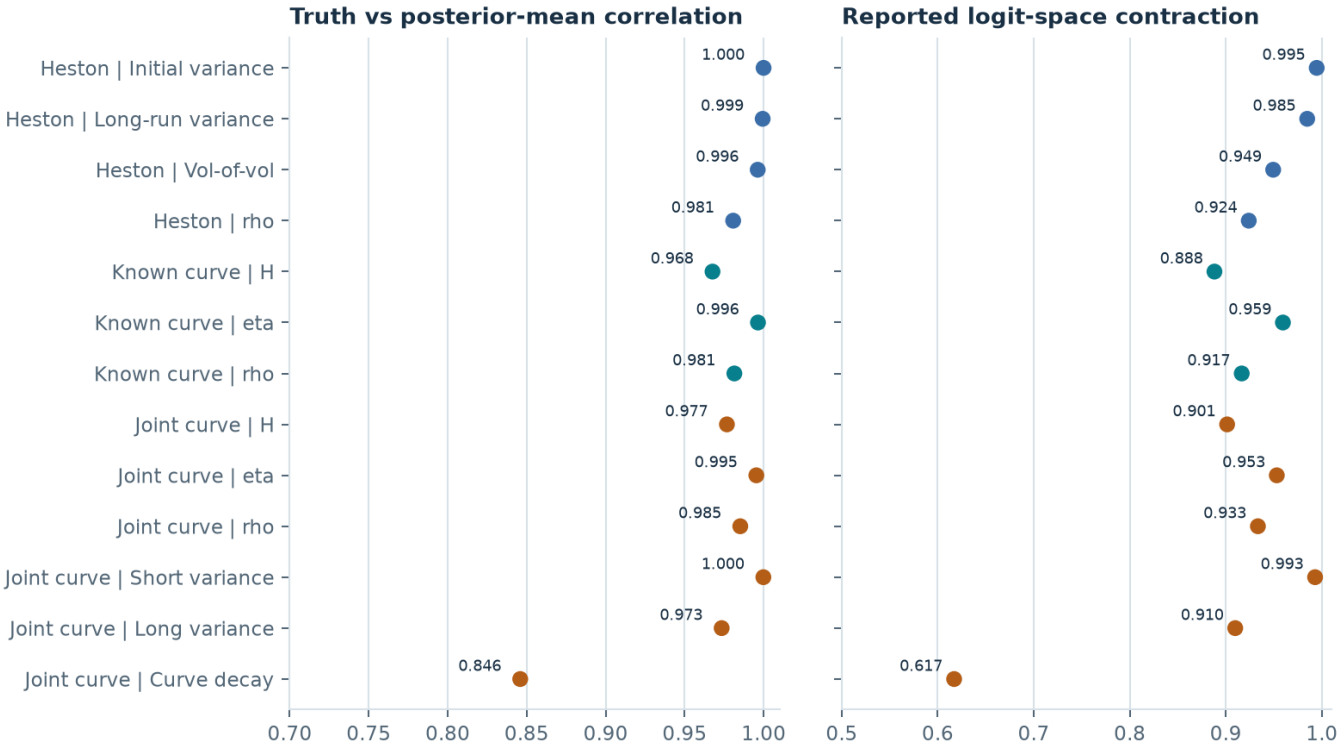
Heston's 50% intervals slightly overcover initial and long-run variance (52.25% / 52.30%). Known-curve rho x eta has 93.85% coverage for 95% intervals (pointwise Wilson upper 94.82%). Joint-curve decay has 46.75% coverage for 50% intervals; short-variance intervals overcover and eta's 95% coverage is 94.00%.

The retained Wilson intervals are pointwise and exploratory, without adjustment across parameters and coverage levels. These qualifications coexist with passing simultaneous rank screens. Neither result proves conditional or full joint-posterior calibration.

03 / IDENTIFICATION

What the observations identify

All three estimators recover substantial signal across the simulated prior. The joint curve's decay parameter is visibly less constrained than its short variance level, an interpretable weak-identification direction rather than an execution failure.



Left: correlation between generating parameters and posterior means across the evaluation panel. Right: the evaluator's logit-space contraction statistic; higher values indicate a larger posterior variance reduction relative to the transformed uniform prior. All primary inferred parameters are shown. Derived products are omitted from this contraction calculation. [S1-S3]

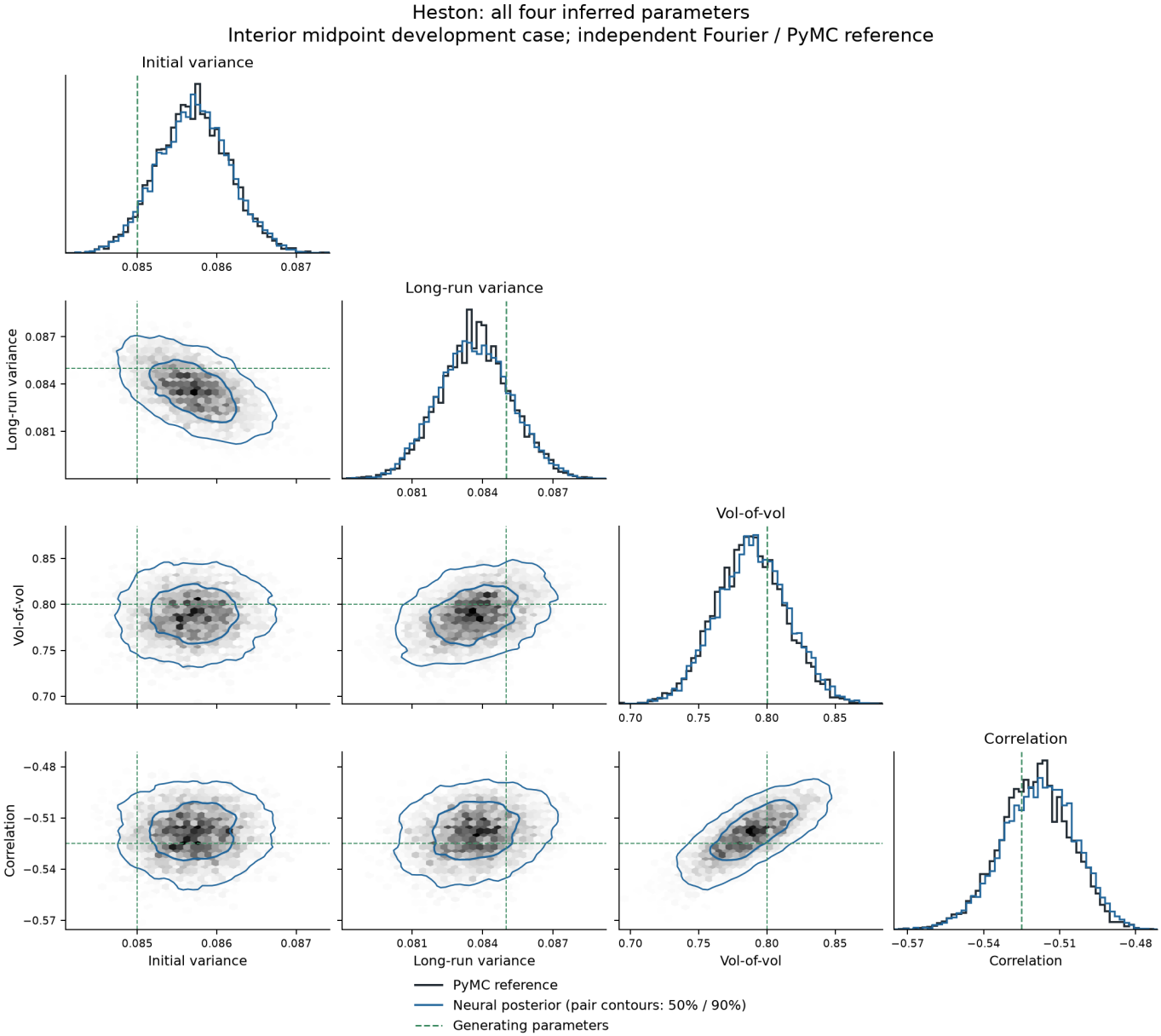
Scenario	Inference target	Legitimate observed context
Heston	Initial variance, long-run variance, vol-of-vol, rho	The fixed complete contract design; kappa = 2
Known-curve rough	H, eta, rho	Seven known variance-curve levels vary between cases and are fixed within each inference
Joint-curve rough	H, eta, rho, short variance, long variance, decay	Contracts and quotes; the generating curve must not enter the conditioning input

All models use BayesFlow 2.1 FlowMatching with vector conditions and four 256-unit layers, final epoch 500, and its native adaptive solver. These results do not yet compare affine/spline flows, MeanFlow, DeepSet or SetTransformer. Their comparisons are independent follow-on work, not an explanation already established by these runs.

04 / INDEPENDENT HESTON REFERENCE

A complete interior posterior

This four-dimensional development case compares the same Heston 100k final-500 candidate against an independent Fourier-likelihood PyMC reference. All inferred parameters and every pair are visible; the reference's central 99% intervals lie comfortably inside the prior.



Black/gray: four PyMC DEMetropolisZ chains, 8,000 retained draws each. Blue: 10,000 neural posterior draws. Green dashed lines: generating parameters. This example is mostly regular; it is not evidence of multimodality or a dramatic banana. [S5,S9]

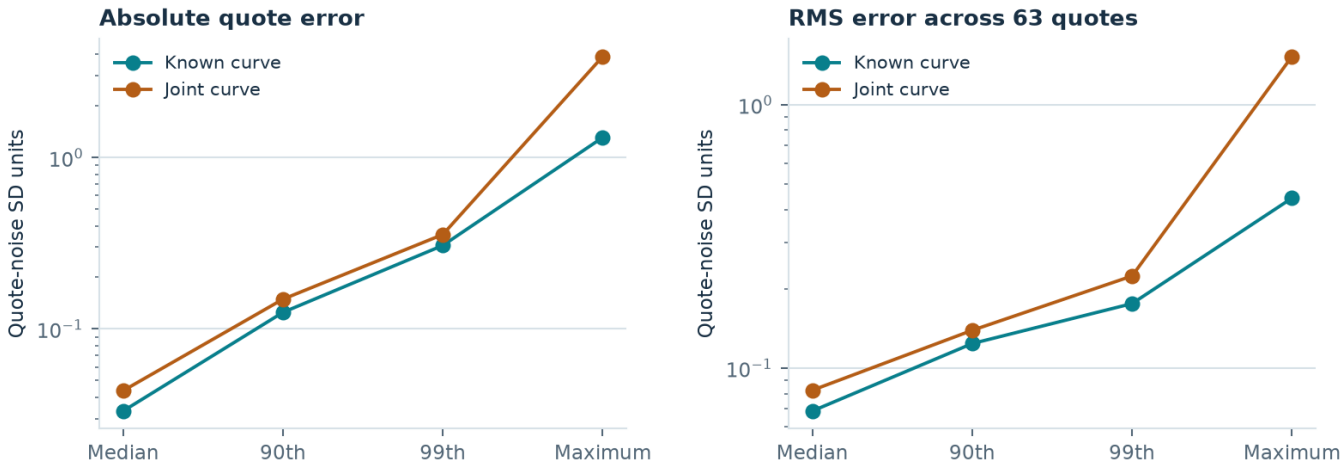
Reference / precision	Neural comparison
Max R-hat 1.00297; min bulk ESS 1,908; min tail ESS 2,607. Largest tested Fourier price shift: 2.97e-6 quote SD.	Max mean discrepancy 0.1013 reference SD; 90% width ratios 1.016-1.040. Largest marginal CDF discrepancy 0.064 (rho).

An ex-ante midpoint case and descriptive development comparison, not another untouched conditional acceptance collection. Earlier boundary examples and their failures remain preserved.

05 / PRICING AND PRODUCTS

Where neural inference meets prices

The forward emulators approximate the numerical bank's conditional mean prices. Validation interpolation error mixes approximation error with label randomness; it does not measure continuous-model bias or validate an emulator likelihood.



Current known 50k: RMSE 0.08287 quote SD, worst quote 1.311 SD. Current joint 20k: RMSE 0.10118, worst quote 3.888 SD and worst surface RMS 1.522 SD. Rare large errors matter for likelihood products across 63 quotes. [S6,S7]

New: the current 50k emulator improves the independent benchmark

Fixed point	Old 20k coarse RMS	Current 50k coarse RMS	Current 50k refined RMS
Case 0 / generating vector	0.0817	0.0381	0.0535
Case 0 / old neural mean	0.0763	0.0359	0.0527
Case 0 / old PyMC mean	0.0772	0.0363	0.0526
Case 2 / generating vector	0.2536	0.1052	0.1641
Case 2 / old neural mean	0.1658	0.0519	0.1285
Case 2 / old PyMC mean	0.1946	0.0743	0.1390

All six current-emulator coarse-grid RMS errors are smaller at these fixed benchmark points. This rescore reused 48 saved independent batches with no new paths or MCMC. The neural/PyMC vectors belong to the earlier 20k study, not current 50k posterior means. RMS entries are point estimates in quote-noise SD units; the source retains per-quote t7 intervals and replicate covariance. Separate new PyMC results follow on page 6. [S10]

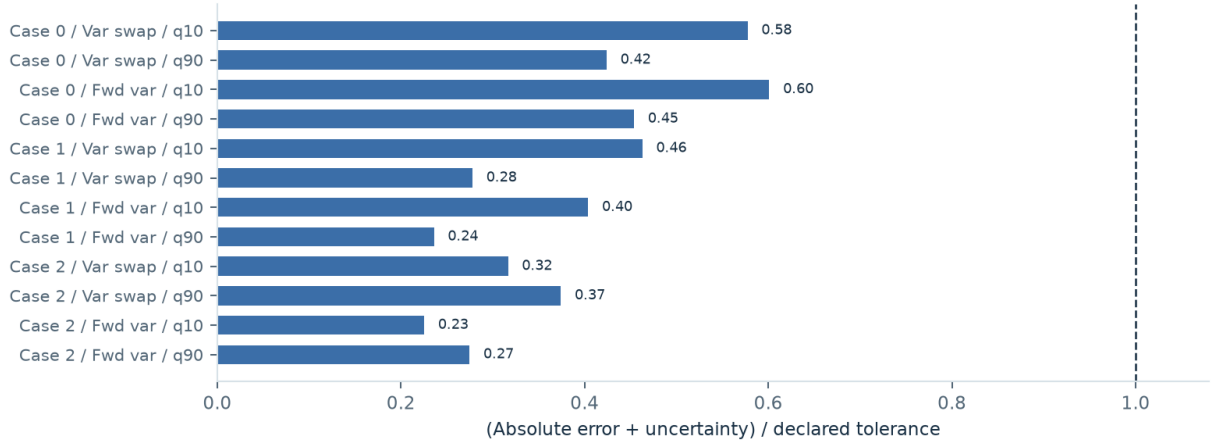
Refined-grid RMS remains 0.0526-0.1641 and the largest inspected refined-quote error is 0.5683 SD. The refined grid is a comparator, not exact truth. Grid shifts are unchanged by retraining; these six points cannot establish posterior-region or prior-wide accuracy, or attribute improvement solely to training-data count.

06 / REFERENCES AND PRODUCTS

Evidence beyond marginal calibration

Heston has an independent Fourier-likelihood PyMC comparison and checks on propagated variance uncertainty. A new known-curve rough PyMC run improves agreement for its approximate emulator target; agreement with the full numerical model remains unfinished.

Heston analytic product uncertainty



All 12 development-product q10/q90 comparisons are inside the declared tolerance including chain-aware quantile uncertainty. The products are the one-year variance-swap rate and six-to-twelve-month forward variance. They test variance-curve uncertainty, not leverage, vol-of-vol or path-dependent payoffs. [S4]

New: current known-curve 50k PyMC comparison

Four native PyMC DEMetropolisZ chains x 8,000 retained draws per case pass all three chain/derived diagnostic screens. With the same observations and known curves, parameter checks pass cases 0 and 1 (2/3 versus old 1/3), using 0.3-SD mean and 25% relative-width rules. [S13,S14]

Case 2 diagnostic	Mean error / reference SD	Neural / PyMC 90% width
H	0.822	1.814
eta	0.789	1.318
rho	0.290	1.423
rho x eta*	0.103	1.451

Case 2 still fails and lies near the eta prior cap: its eta mean discrepancy improves from 2.052 to 0.789 SD, while intervals remain too wide. Current comparisons reuse 1,000 saved neural draws versus 10,000 previously; neural Monte Carlo uncertainty and uncomputed reference-quantile MCSE qualify the width screens. Improvement cannot be attributed solely to more training data. [S8,S13,S14]

A converged emulator-PyMC chain is a reference for that emulator target. It is not automatically a reference for the stochastic bank simulator or continuous-time model: numerical grid bias, emulator approximation and Monte Carlo label variability remain separate errors.

What remains for the rough references

Joint 20k has no completed conditional PyMC comparison. Ten local analytic/finite-difference/compiled PyTensor gradient checks pass; gradients alone do not establish likelihood or posterior accuracy. Next: independent prices around posterior mass, then qualified conditional and product comparisons. [S10,S12,S14]

Scientific-acceptance flags remain false; reserved rough inference-test outcomes were not opened.

07 / EVIDENCE AND NEXT DECISIONS

What is ready, and what comes next

Usable substrate: three baseline estimators with offline replays, calibration evidence, an independently checked Heston example, and a later known-curve 100k fit. Remaining work concerns conditional rough inference, pricing tails, discretization, products and a full-100k joint fit.

Parallel priorities

Review the completed known-curve 100k fit before promotion. The matched comparison on page 8 separates posterior behavior from a worse new price emulator. Preserve the 50k cache/reference and Heston's frozen test boundary.

Resolve the rough reference limitations. Preserve the new known-curve 2/3 result and case-2 failure; check independent prices around posterior mass and prepare a joint-curve reference. Emulator agreement is one development screen. [S13,S14]

Prepare comparable architecture and summary experiments. Separate inference-family changes from summary-network changes; match data, target, seeds and measured compute, and retain failed runs.

Review the prepared offline backbone notebook. The new backbone_review.py replays pinned Heston, known 50k and joint 20k artifacts; it was statically checked, not executed. Cached examples are not fresh inference or new acceptance, and the joint fixture has prior-edge proximity. [S11]

Model identities

Heston / 100,000 training surfaces / checkpoint `8d4a243ba099524cf4cda40e4f0616e895879f053829a7f60dadbe65b98301db`

Rough Bergomi / known curve / 50,000 training surfaces / checkpoint `e32732daea513c1887438eab801e660f286166846ef65eac6258136c3f504613`

Rough Bergomi / joint curve / 20,000 training surfaces / checkpoint `8d552a9392d54f98a954401fb979e861f26b9a9c71690b5c3980a4e8fa608dec`

Reproducible evidence

[S1] Heston frozen-candidate held-out evaluation

[S2] Known-curve 50k validation

[S3] Joint-curve 20k validation

[S4] Heston analytic product comparisons

[S5] Heston interior PyMC / neural verification

[S6] Known-curve 50k pricing-emulator report

[S7] Joint-curve 20k pricing-emulator report

[S8] Earlier known-curve independent pricing checks (20k model)

[S9] Complete Heston interior corner provenance

[S10] Known 50k fixed-point rescore and compiled gradient checks

[S11] Pinned offline backbone artifact catalogue

[S12] Joint 20k compiled gradient checks

[S13] Current known 50k PyMC completion and fixed-case comparisons

[S14] Current known 50k reference target and Monte Carlo qualifications

[S15] Matched known-curve 50k versus 100k learning comparison

[S16] Unchanged validation data and new model identity proof

[S17] 100k pricing-emulator independent fixed-point rescore

[S18] 50k and 100k neural estimators against fixed 50k PyMC chains

[S19] Completed known-curve 100k validation evaluation

The companion Markdown lists exact repository paths and full file hashes. Re-render with scripts/render_inference_performance_report.py. The renderer reads retained JSON and one existing figure; it performs no simulation, training, MCMC, notebook execution or new test evaluation.

08 / MATCHED LEARNING COMPARISON

More data: what actually changed

The known-curve 100k fit completed. Marginal calibration changes little; some conditional discrepancies shrink, but the new price emulator is worse. Bank growth is useful infrastructure, not a guarantee that every fitted component improves.

Same 2,000 validation cases x 1,000 draws; all 5,000 validation surfaces and the original 50k training prefix have identical hashes. Both fits use seed 0, the same architecture, default adaptive sampler and final epoch 500. Updates differ: 98,000 versus 195,500. This is one seed, not a replicated or fixed-compute learning curve. [S15,S16]

Diagnostic target	Rank KS: 50k / 100k	90% coverage: 50k / 100k
H	0.02276 / 0.02121	89.30% / 89.45%
eta	0.02376 / 0.02199	89.45% / 89.40%
rho	0.02194 / 0.01840	89.85% / 89.85%
rho x eta*	0.01562 / 0.01708	89.70% / 89.90%

Both four-target marginal rank families pass the same 0.03562 limit; rho x eta is derived downstream. Small KS changes and nearly unchanged coverage do not establish a material posterior improvement. [S15,S19]

The 100k price emulator regresses

Validation label metric	50k pricer	100k pricer
RMSE / quote-noise SD	0.08287	0.09843
Epochs run / selected	300 / 298	116 / 86

Label RMSE worsens 18.8%, and independent coarse/refined RMS errors rise at all six reused benchmark points. Case-2 truth coarse RMS: 0.1052 to 0.2426 quote SD. Early stopping selected very different training stages: an optimization confound, not a proven explanation or evidence that more data is harmful. [S15,S17]

Hold the 50k PyMC reference fixed

Case 2 target	Mean error / SD: 50k / 100k	90% width ratio: 50k / 100k
H	0.822 / 0.565	1.814 / 1.587
eta	0.789 / 0.662	1.318 / 1.293
rho	0.290 / 0.103	1.423 / 1.169
rho x eta*	0.103 / 0.259	1.451 / 1.188

Both saved neural models pass 2/3 fixed-case parameter screens; case 2 still fails in H and eta. Each uses 1,000 draws against exactly the same four x 8,000 50k-emulator PyMC draws, observations and curves. No new MCMC; the worse 100k pricer is not the reference. Quantile/neural Monte Carlo errors remain unassessed; this approximate-target comparison is not acceptance. The 50k cache stays unchanged. [S18]

New 100k posterior: `39f1a81986a38994dc6eacc520ad95254b34346abc02e758c08779c7c5916fc4`